What this review needs to decide

A useful part for the expert visit

The owner needs an accurate, simple backing-plate design that can be quoted and physically tried soon. Pro should issue concrete instructions to the local agent for revised CAD and separate fabricator, repair and installation handoffs.

Who measures and assembles

The expert is expected to assemble/install, not supply new CAD measurements. Measuring was difficult for the owner. Ask the owner for at most a handful of indispensable machining dimensions, only after using existing evidence and simpler design options. Each request needs the affected CAD decision, illustrated contacts and plane, easiest method and necessary precision. General field forms are optional; ordinary installation checks remain separate.

Current proposal to challenge

Main: diameter 220 mm, centered upright capsule 116.35 x 151.10 mm, end radius 58.175 mm. Diverter: 124 mm outside / 80 mm inside; 120 mm fallback. Stock: 1.524 mm metal or matching plastic; separate 3 mm plastic trial. No locators in the present files.

Four decisions

Choose the best aperture while preserving full foam support and real control access. Decide whether bolt notches or other locators are useful and feasible. Establish the minimum factory-fit and water-path checks for the expert. Specify the documents and file-level tests the agent should produce.

Read the pictures according to their labels

Inner-plate face dimensions were measured within one plane; the owner considers those caliper readings highly accurate. Installed tip distances span unequal depths and are 3D lengths. Proposed CAD and exploded pictures contain assumptions. GP03 clarifies this distinction; FT01 CAD is unchanged.

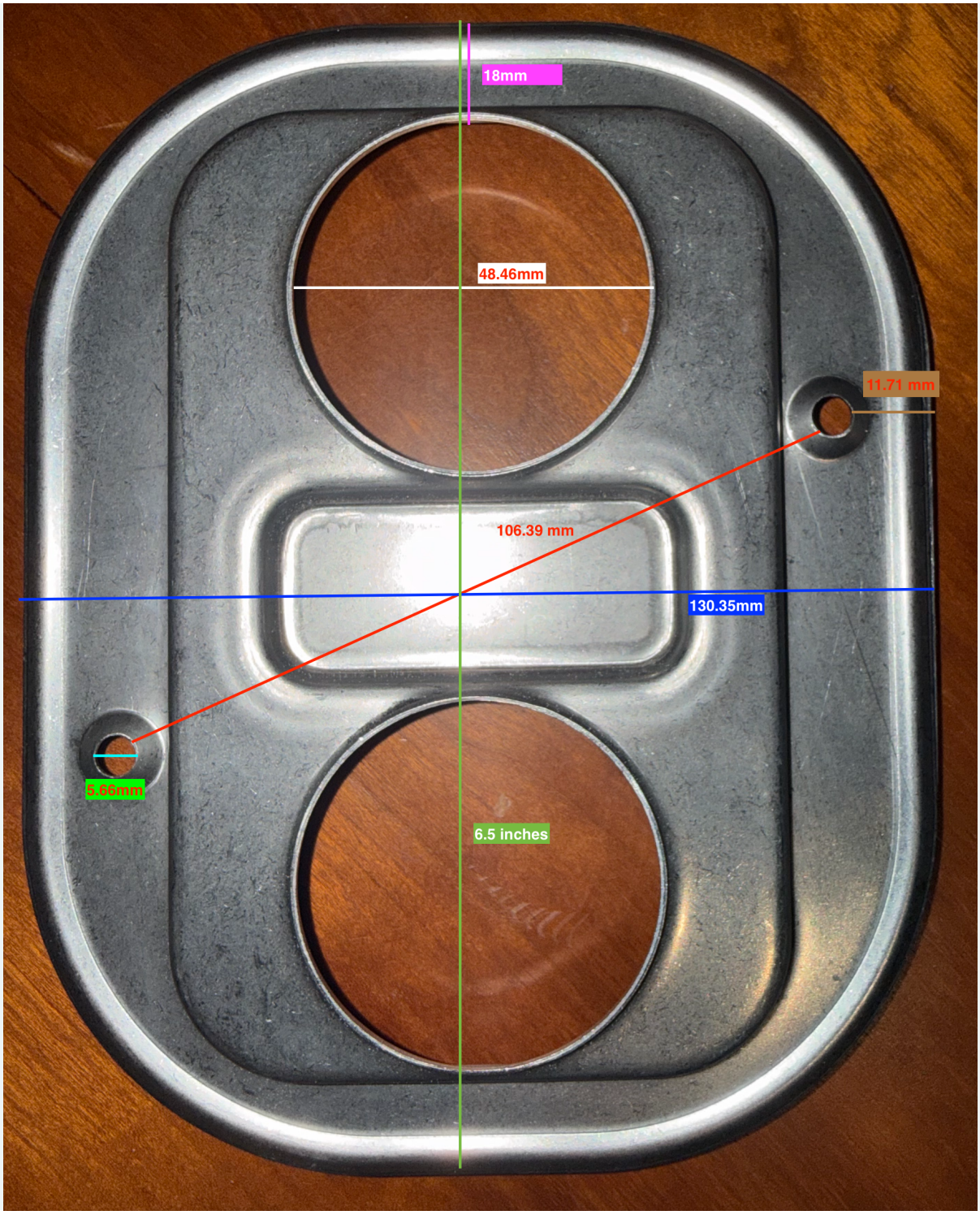
What has not been established

Physical handle attachment, compressed foam contact, sound support and retention, the final sealing detail, actual paired gap, and installed water performance remain open. The existing digital audit does not certify these. The nominal 1 mm extra foam land is particularly narrow.

Use the separate text prompt

01_PASTE_INTO_GPT_PRO.txt asks for a decision and a self-contained BEGIN AGENT INSTRUCTIONS block. 02_PROJECT_BRIEF.txt holds the full measured facts and current limitations. The visual pages supplement those files; they do not replace them.

Inner plate measured face

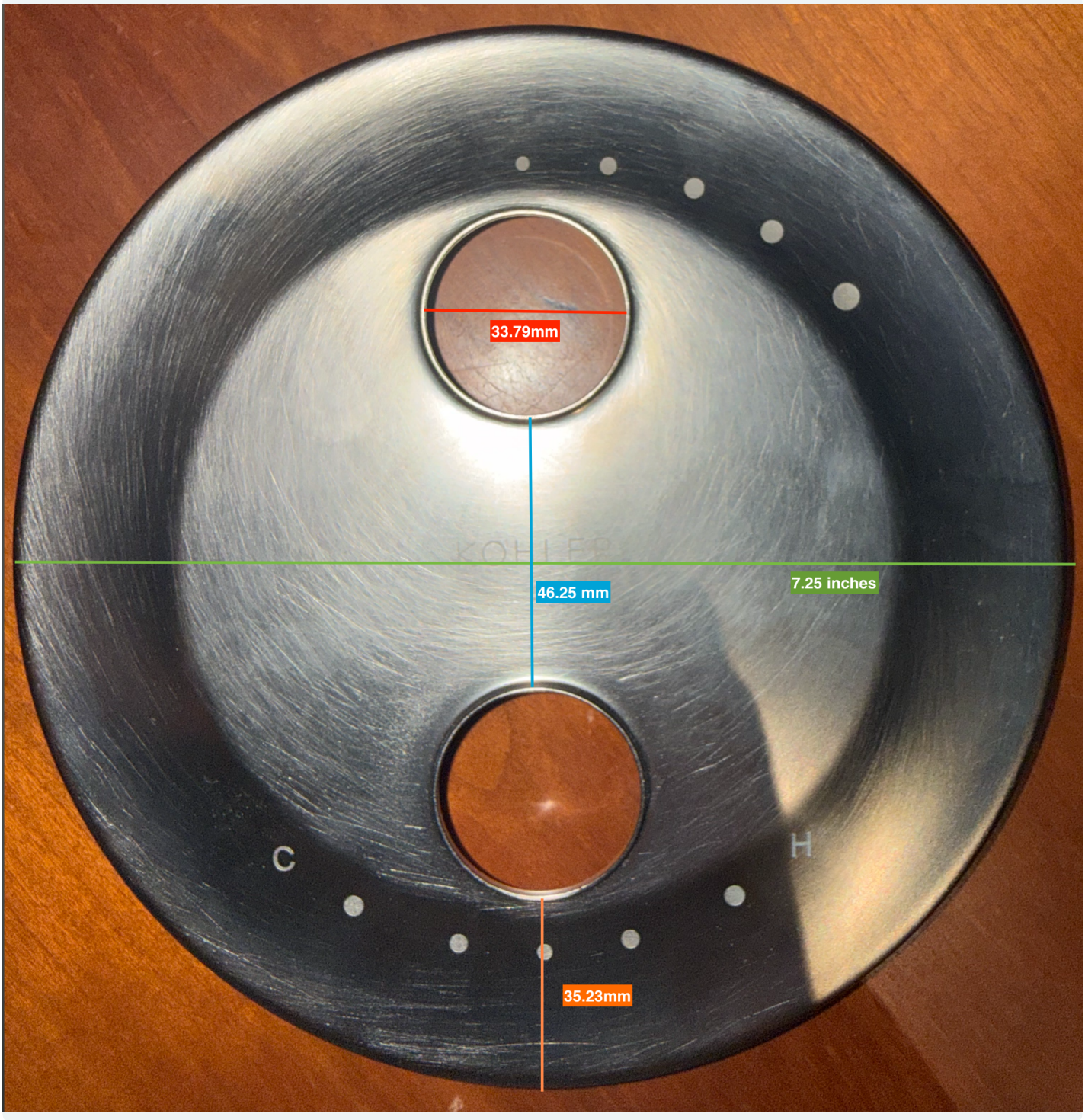


Measured within one plane; the owner considers the caliper readings highly accurate. No tip-depth correction applies. Width 130.35 mm; upper opening 48.46 mm; tape height 165.10 mm.

106.39 mm is nearest-hole-edge gap; 11.71 mm is hole edge to outside rim; SW hole is 5.66 mm; NE diameter is unknown. Use the calipers ahead of scan estimates for these measured dimensions. Foam/depth readings are separate; no numerical accuracy bound was supplied.

Source: data/evidence/user/inner-plate-front-caliper-2026-10-14.png

Decorative cover measured face



Outside diameter 7.25 inches = 184.15 mm. Upper opening 33.79 mm. The blue 46.25 mm is between nearest inside opening edges. Orange 35.23 mm runs from lower inside opening edge to outside bottom rim.

Lower opening diameter and rear cavity are unmeasured. These front dimensions do not establish nut clearance, handle engagement or the installed center.

Source: data/evidence/user/outer-plate-front-caliper-2026-10-14.png

Upper-right stud tip distances

U-NE = 51.40 mm; L-NE = 80.25 mm. These are 3D distances between end-tip centers at different depths. NE is the upper-right trim rod. Unequal depth makes a tip distance longer than its projection onto the tile plane.

Additional supplied lengths: U-L = 80.00 mm and NE-SW = 109.80 mm. Do not use these directly as CAD hole spacing. Unlike the coplanar inner-plate face measurements on page 2, projected tip spacing needs the depth difference; stud axes at the backing plane remain separate.

Source: data/evidence/user/installed-ne-stud-distances-2026-10-15.png

Lower-left stud tip distances

U-SW = 81.14 mm; L-SW = 54.70 mm. These join end-tip centers at different depths, not points in the tile plane. The 3D lengths exceed their tile-plane projections. SW is the lower-left trim rod.

Do not use these directly as CAD hole spacing. The inner-plate face measurements on page 2 are coplanar and need no tip-depth correction. Here, depth differences are needed for projection; stud inclination and axis intersections at the backing plane are still unestablished.

Source: data/evidence/user/installed-sw-stud-distances-2026-10-15.png

ACTUAL USER PHOTOGRAPH / NO CAD RECONSTRUCTION

Rear installation context

Actual rear view of wood, valve and plumbing. The foam discussed elsewhere belongs to the removable front inner plate; no foam-filled rear cavity is claimed.

The owner confirms rear rod contact with wood and conditionally prefers clearance behind the rods. This photograph does not supply a drilling depth, member classification or a clear route.

Source: data/evidence/user/rear-context-2026-10-15.jpeg

ACTUAL USER PHOTOGRAPH / OWNER-CONFIRMED CONTACT

Rear rod and wood relationship

The owner confirms the nominal 4.56 inch trim rods meet the wood. The desired remedy is inward trim-stud adjustment where feasible, after checking the other valve and handle depths.

Nominal full rod length 115.824 mm is not exposed projection. The older guide gives 11.1125 mm maximum front projection from the finished wall; applicability and actual datums require confirmation. Do not convert this image into a wood drilling specification.

Source: data/evidence/user/rear-close-2026-10-15.jpeg

Depth concern from the side

The projecting control tips and trim rods end at different Z depths. The original low-resolution side view is retained without inventing hidden geometry.

Handle connection depth and trim-stud projection are separate constraints. The custom backing adds actual stock thickness plus effective bedding height. There is no established 5 mm engagement or waterproofing allowance.

Source: data/evidence/user/depth-side-view-2026-10-14.png

Real inner plate attached

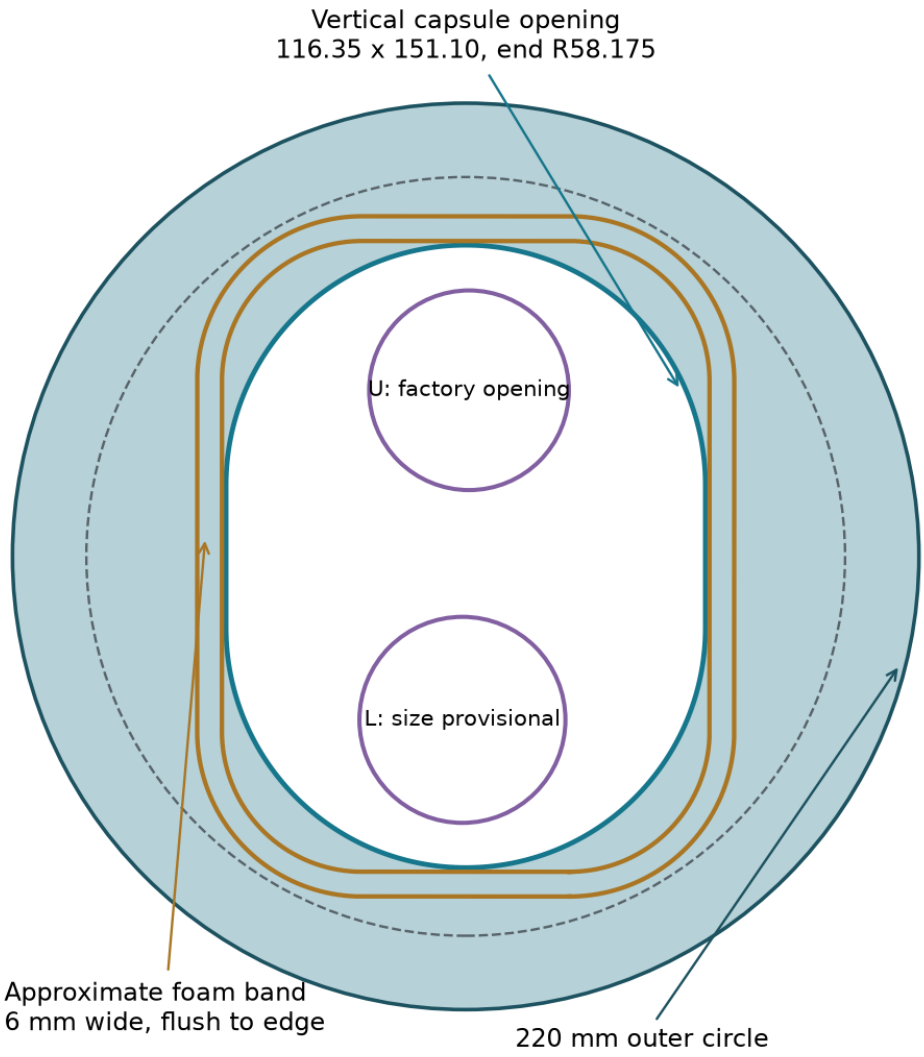
The real attached-plate capture shows stem centers apparently high inside visible hub rings. Approximate offsets are 3-4 mm in this inspection, not proven axis errors. Do not force visual concentricity.

Scale was calibrated to the supplied 80 mm 3D U-L length. Outline width reads about 134.46 mm versus 130.35 mm caliper. This scan is not globally registered to the primary capture and cannot serve as a precision mounting jig.

Source: FT01/plane_front.png

Current main aperture

Custom opening and provisional factory opening footprints



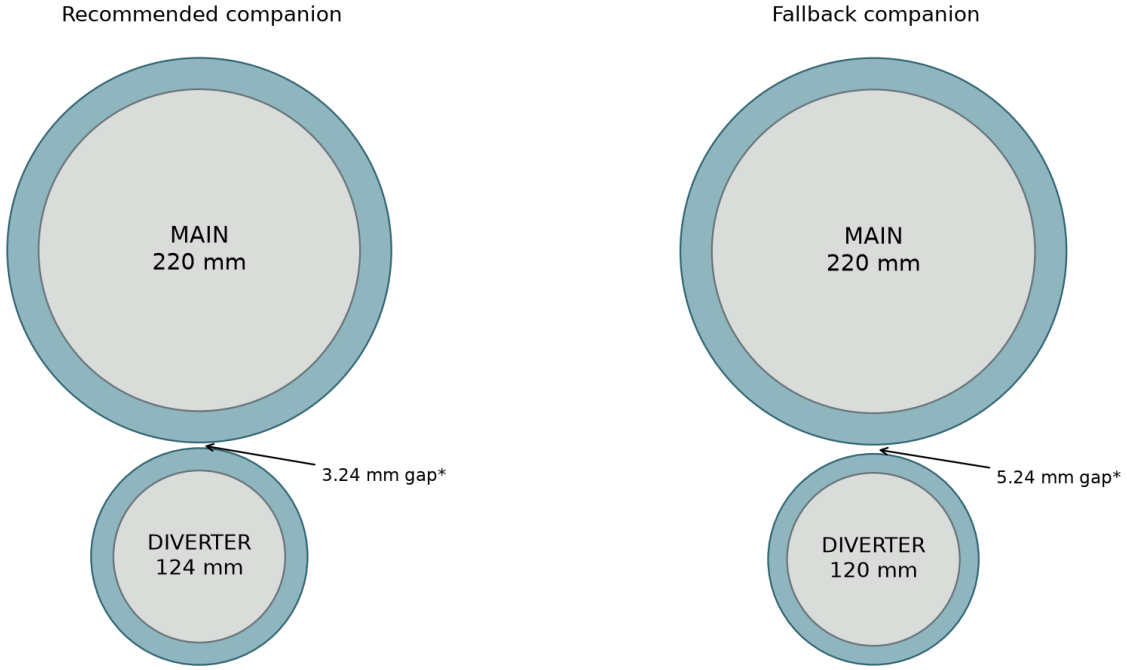
Gold is a nominal footprint, not a verified seal. 1.00 mm extra land inside the assumed foam.
Purple circles belong to the factory inner plate; they are not extra holes cut in this part.

Main outside diameter 220 mm; centered vertical capsule 116.35 x 151.10 mm with R58.175 ends. Gold represents assumed edge-flush foam. The lower opening circle and some placement dimensions remain assumptions.

The nominal extra land inside the 6 mm foam is only 1 mm. The smallest scoped projected hardware clearance is about 1.61 mm before unresolved scan/placement errors. These are review targets, not accepted field tolerances.

Source: FT01/access.png

Main and diverter size choice



*Original placement estimate. Across recorded scenarios, selected-pair gap is 3.06-5.21 mm, not error bounds. Increasing either diameter consumes half that increase from the gap. Verify with both real covers.

Recommended trial: main 220 mm and diverter 124/80 mm. The separate diverter fallback is 120/80 mm. Keep both backing circles concentric with their actual covers.

Earlier placements predict only about 3-5 mm between the larger pair. These are scenarios, not an uncertainty range. The actual diverter seal footprint and paired clearance need the expert fit trial.

Source: FT01/sizes.png

Proposed appearance on the travertine

Illustrative exterior appearance of both backing plates together on the primary captured travertine. Factory exterior forms are based on official CAD.

Rendered finish, contact, engagement and sealing are not physical results. This image is useful for appearance; the critical hidden interfaces must be evaluated separately.

Source: FT01/paired_front.png

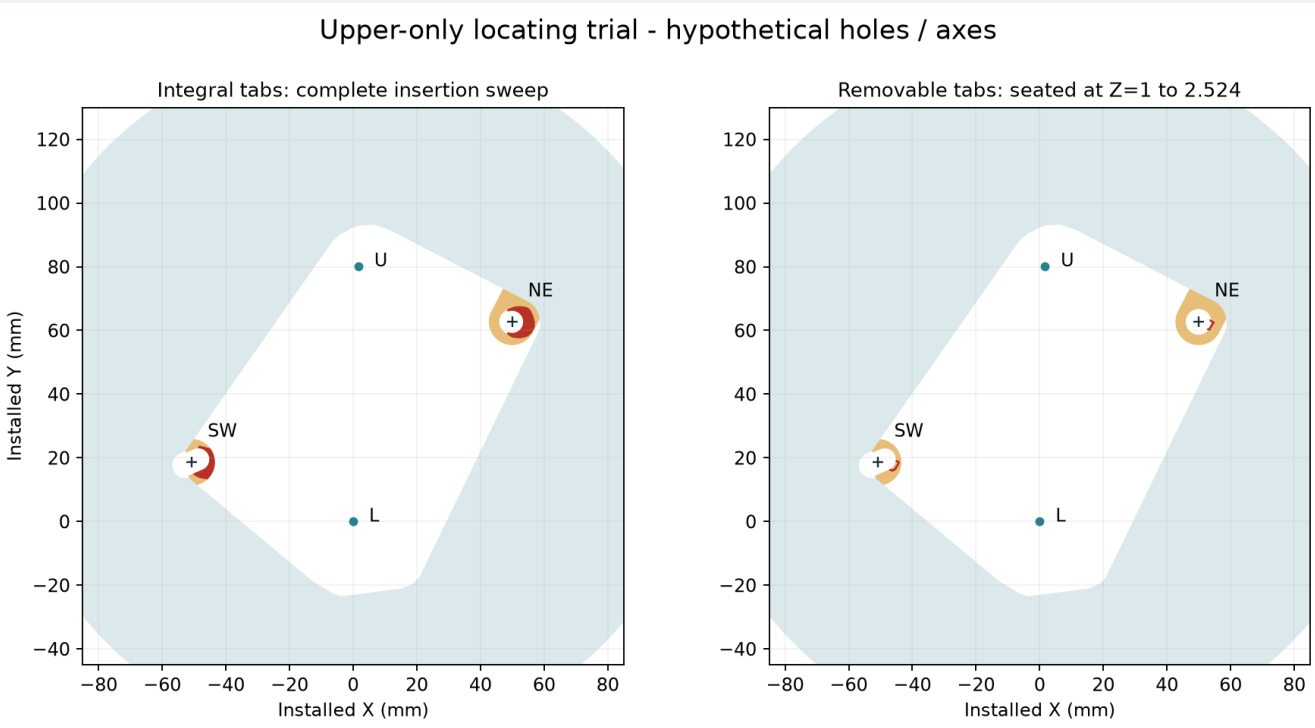
Exploded layer illustration

Wall, proposed backing, factory inner plate/foam, cover and controls are separated for viewing. The real foam is attached to the inner plate. Display spacing is not installed spacing.

Rods and nuts are omitted from this appearance view; it is not an assembly-path proof. The requested final schematics must include their actual envelopes behind the unperforated decorative cover and section both control connections.

Source: FT01/exploded.png

Earlier locating-tab experiment



Amber: added tab metal. Red: projected captured-surface interference. Not a released mounting design.

This earlier diagonal opening added metal tabs with hypothetical holes/axes. Red marks conditional captured-surface interference in insertion or seated trials. It is not the FT01 capsule design.

The trial does not rule out every notch. A notch removes material; a tab adds it. Pro should independently assess restraint, adjustment, foam land, bearing, depth and complete insertion/removal before recommending a locator variant.

Source: deliverables/current/fabrication/concentric-border-study/views/stud_locator_trial.png